

DINESH NATARAJAN

Aerospace Engineer | FEA · CFD · Flight Dynamics · GD&T · V&V

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PROFESSIONAL SUMMARY

Mechanical design engineer (MSc Aerospace, University of Salford) specialising in 3D modelling, detailed 2D drawings and GD&T (ASME Y14.5-2018), backed by structural FEA (ANSYS), tolerance stack-up analysis and Siemens Teamcenter PLM. Pairs a design-for-manufacture mindset with applied machine learning for FEA surrogate modelling.

TECHNICAL SKILLS

CAD & Drawing: CATIA V5, Creo (Pro/ENGINEER), SolidWorks - 3D solid/assembly modelling, detailed 2D drawings, machined-part design

Design & Analysis: GD&T (ASME Y14.5-2018), Tolerance stack-up (worst-case / RSS), Design-for-Manufacture (DFM) & Assembly (DFA), material selection, Structural FEA & Topology Optimisation (ANSYS)

PLM & Process: Siemens Teamcenter (item & revision management, EBOM, dataset management, workflow-based drawing release, engineering change), V-Model, V&V, MIL-SPEC compliance, Requirements analysis; MATLAB / Simulink, Python

PROFESSIONAL EXPERIENCE

Product Design Engineer — Intern

Oct 2020 - Mar 2021

Black Space Research & Development, Bengaluru

- Performed linear static FEA on a Rear Underrun Protection Device (RUPD), assessing structural strength under regulatory load cases and feeding results into design refinement.
- Produced 3D CAD models and detailed 2D part/assembly drawings, collaborating cross-functionally to mature manufacturable, requirement-compliant designs.

KEY ENGINEERING PROJECTS

Aircraft Engine Mount Bracket — Mechanical Design, GD&T & Tolerance Analysis

University of Salford (MSc) · CATIA V5 · ANSYS Mechanical · ASME Y14.5-2018

- Produced the 3D solid model and a fully annotated 2D GD&T drawing of a machined aerospace bracket (SAE 4130), defining a function-driven A|B|C datum scheme per ASME Y14.5-2018.
- Managed the complete part lifecycle in Siemens Teamcenter (PLM) - item & revision control, EBOM structure, dataset management and workflow-based drawing release - delivering controlled, release-ready engineering data.
- Applied GD&T and 1D tolerance stack-up analysis (worst-case + RSS) plus a floating-fastener / MMC study to prove a manufacturable, standards-compliant bolted interface.
- Drove a production-ready design via structural FEA across 6+ load cases and topology optimisation (~22% mass reduction); material selection (SAE 4130) and DFM/DFA kept all results within aerospace limits across static, vibration and thermal conditions.

Machine-Learning Surrogate Model for FEA — Engine Bracket Extension

Python (scikit-learn) · surrogate / regression modelling · ANSYS dataset

- Trained a machine-learning surrogate on the bracket FEA dataset to predict equivalent stress, deformation and mass from design inputs - returning near-instant estimates in place of full solver runs.
- Enabled rapid design-space exploration and lightweighting trade studies, linking the GD&T / FEA work to a data-driven optimisation workflow.

Aircraft Flight Dynamics Simulation — Cessna 172

MATLAB / Simulink · MIL-SPEC

- Developed a 4-DOF state-space flight-dynamics model; analysed longitudinal/lateral stability and dynamic response, verified against MIL-SPEC handling-qualities criteria.

Natural-Laminar-Flow Airfoil CFD Analysis

XFLR5

- Compared NACA 2412 vs NACA 63-215 across the cruise envelope; achieved ~44% lift-to-drag improvement and ~42% drag reduction at cruise through NLF airfoil selection.

Twin-Boom UAV — Conceptual Design & Aerodynamic Analysis

University of Salford (MSc) · SolidWorks · XFLR5

- Designed a twin-boom UAV configuration; modelled the airframe in SolidWorks and evaluated aerodynamic performance and stability in XFLR5.

EDUCATION

MSc Aerospace Engineering

2021 - 2023

University of Salford, Manchester, UK — Merit (2:1)

BTech Aerospace Engineering

2017 - 2021

Karunya Institute of Technology & Sciences, Coimbatore — First Class (2:1)

CERTIFICATIONS

- Master Diploma in Product Design** — CADD Centre (Completed, Jan 2022)
- Airbus Supply Chain Specialist** — Airbus Beyond (Completed, May 2026)
- Machine Learning Specialization** — DeepLearning.AI & Stanford Online (In Progress, expected 2026)